DRAFT APPENDIX FOR
AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT
HISTORICAL NOTE: INTEGRAL CAYLEY NUMBERS, 1923–1946

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ABSTRACT. This is a draft of the historical appendix planned for v4. It is written *solely from the primary sources*: the four papers Dickson (1923), Kirmse (1924), Mahler (1942), and Coxeter (1946), each obtained and re-derived in exact arithmetic in this project (`python_project/src/verify_dickson_1923.py`, `verify_kirmse_1924.py`, `verify_mahler_1942.py`, `verify_coxeter_1946.py`; companion notes in `paper/reviews/`). No secondary source is taken on trust.

SETTING

The integral-Cayley-number construction – a maximal order in the real octonion algebra, supported on the E_8 lattice – was developed in four papers between 1923 and 1946. Each contributes one piece. Together they fix three facts that underlie the present construction: there are *seven* maximal orders; each is obtained from any one by a transposition of two imaginary basis units (a linear involution of $\mathbb{R}^8$); and the underlying lattice is E_8. The third fact and the count are already in Dickson; the second is the Coxeter–Bruck remedy of Kirmse.

DICKSON 1923

L. E. Dickson [Dickson1923], in Section 19 of *A new simple theory of hypercomplex integers*, gave the first construction. He built the octonions by Cayley–Dickson doubling of the quaternions, $x = q + Qe$ with q, Q quaternions, and the product (his equation (24))

$$(q + Qe)(r + Re) \;=\; (qr - \bar{R}Q) + (Rq + Q\bar{r})\,e,$$

and then determined the maximal sets of integers closed under his three axioms – closure C, contains unit U, integral norm N (his Section 6). From the norm form he derived that integer coordinates are either integers or half-integers, ran a case analysis, and exhibited (his (29)) the basis

$$i,\; j,\; k,\; p,\; e,\; W,\; Z,\; V, \qquad p = \tfrac{1}{2}(1 + i + j + k),$$

of an order, his *system (30)* (here denoted O_D). All 64 products of the basis (29) lie in O_D; the system is genuinely closed [Dickson1923].

Dickson then stated as **Theorem XV**:

> *The only maximal systems of integers of Cayley's algebra having properties C, U, N are (30) and the two systems obtained from it by cyclic permutations of i, j, k.*

This undercounts. Re-derived in this project, the true number of maximal orders for Dickson's product is *seven*, not three. The cause is in Dickson's own derivation: on p. 321, having reached a case in which the order contains the Hurwitz quaternions $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$, he writes

> *"... and hence all of Hurwitz's integral quaternions. We shall assume that the latter occur in our set in all cases."*

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The assumption that every maximal order contains the Hurwitz quaternions is stronger than C, U, N. Of the seven maximal orders, *three* contain those quaternions, and those three are precisely O_D and its two $(i\,j\,k)$-cyclic images – exactly Dickson's Theorem XV triple. The four he missed are exactly the maximal orders that do not contain the integral quaternions.

So Dickson's *construction* is correct; his *count* is too small by four, by a qualifier that he stated openly in the derivation but that was absorbed into the theorem.

Kirmse 1924

J. Kirmse [Kirmse1924], working in Germany the following year, took up the problem with a different presentation. He fixed an octonion multiplication on the basis $\{1, i_1, \ldots, i_7\}$ by seven cyclic Fano triples (his table (1) on p. 64) and sought the maximal sets of octonion integers closed under multiplication. He stated that there are *eight* such sets (*"Man findet acht solche . . . ,"* p. 70) and described one in detail, his system J_1 (p. 70):

$$J_1 \;=\; \big[\, J_0,\ a_1,\ a_2,\ a_3,\ a_4 \,\big],$$

where $J_0 = \mathbb{Z}\langle 1, i_1, \ldots, i_7\rangle$ and

$$a_1 = \tfrac{1}{2}(1 + i_1 + i_2 + i_3), \qquad a_2 = \tfrac{1}{2}(i_4 + i_5 + i_6 + i_7),$$
$$a_3 = \tfrac{1}{2}(1 + i_1 + i_4 + i_5), \qquad a_4 = \tfrac{1}{2}(1 + i_2 + i_4 + i_7).$$

Kirmse's multiplication table is itself a correct octonion algebra: it is norm-multiplicative and satisfies all the Cayley identities. His error is in the lattice J_1: it is *not closed* under his own multiplication. The witness, computed in this project and matching Coxeter (1946) verbatim, is

$$a_1 \cdot a_3 \;=\; \tfrac{1}{2}(1 + i_1 + i_2 + i_3) \cdot \tfrac{1}{2}(1 + i_1 + i_4 + i_5) \;=\; \tfrac{1}{2}(i_1 + i_2 + i_4 + i_7),$$

and the product is not among the 240 elements of J_1 of unit norm (Kirmse p. 76). An enumeration of the 30 candidate E_8-type lattices under Kirmse's table yields exactly *seven* closed ones; J_1 is the spurious entry on Kirmse's list of eight.

So Kirmse's *table* is correct; his *exhibited order* is the defective one, and his *count* is too large by one.

Mahler 1942

K. Mahler [Mahler1942], eighteen years later, worked over a maximal order (one of the seven, to be made explicit by Coxeter in 1946) and developed its arithmetic. His principal results are two.

- **Euclidean property** (his Theorem on p. 127): *for every Cayley number X there is an integral Cayley number G with $N(X - G) \le \tfrac{1}{2}$.* Mahler calls this "not easy to prove." It is sharper than Dickson's earlier 5/4 in place of $\tfrac{1}{2}$.
- **Principal ideals**: *every left (or right) ideal in the ring of integral Cayley numbers is principal, generated by a multiple (in the ordinary sense) of an integral Cayley number of norm* 1*,* 2*, or* 4.

Re-derived in this project, Mahler's results are correct as stated; the printed paper contains one transcription error in equation (10), without consequence for the theorems. Mahler cites Dickson (1923) but not Kirmse; that citation chain is how the priority of Dickson's construction reaches Coxeter in the following paper.

Coxeter 1946

H. S. M. Coxeter [Coxeter1946], in *Integral Cayley numbers*, identifies Kirmse's error and supplies a corrected order. His Section 4 ("Kirmse's mistake") reproduces Kirmse's table as his (4.1) (the two tables are identical, verified) and exhibits the same non-closure witness as above:

$$\tfrac{1}{2}(1 + i_1 + i_2 + i_3) \cdot \tfrac{1}{2}(1 + i_1 + i_4 + i_5) \;=\; \tfrac{1}{2}(i_1 + i_2 + i_4 + i_7),$$

which, Coxeter notes, "is not one of the 240 units." Of the count he writes: *"Actually there are only seven, which presumably are the remaining seven of his eight"* (p. 566).

The remedy is due to **R. H. Bruck**. Coxeter writes: *"When I pointed this out to Bruck, he sent me a revised description of such a domain. Bruck's domain J can be derived from J_1 by transposing two of the i's."* That is, the correction is a transposition of two imaginary basis units of the octonions – a linear involution of $\mathbb{R}^8$. Any of the $\binom{7}{2} = 21$ such transpositions repairs Kirmse's J_1. Coxeter further notes that the 21 "fall into 7 sets of 3, each set having the same effect," yielding the seven maximal orders; this combinatorial structure is verified in this project (each set of 3 is three disjoint transpositions fixing one common imaginary unit, the "special unit" of the corresponding order).

Coxeter then develops the geometry of the corrected order. Its 240 norm-1 units are the vertices of Gosset's polytope 4_{21}; the order itself is the eight-dimensional honeycomb 5_{21}, the densest sphere packing in eight dimensions (his Sections 6–13). He proves the corrected order maximal in his Section 11.

A postscript (Section 14) records that Coxeter learned of Dickson and Mahler only after the manuscript was written, when Olga Taussky Todd drew his attention to Mahler, which in turn cites Dickson. Coxeter there identifies the cause of Dickson's Theorem XV undercount independently:

> *"... he asserted [9; 334] that there are only three systems which include all the eight basic units ... He misused the remaining four of the seven systems by adding the very artificial requirement that the integral Cayley numbers should include the integral quaternions $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$."*

And of Mahler's "not easy to prove" Euclidean theorem, Coxeter observes that it is an immediate consequence of his own Section 11. By 1946 the construction is complete.

Synthesis

The four papers together fix the construction the present paper rests on:

- **1923 (Dickson).** An integral-Cayley-number maximal order exists; the lattice is E_8; the count, in the form Dickson's theorem gives, is three (under his stated extra assumption).
- **1924 (Kirmse).** An independent treatment; correct multiplication; an enumerated count of eight that includes one non-closed candidate.
- **1942 (Mahler).** Arithmetic of the order: Euclidean with $N(X - G) \leq \frac{1}{2}$; principal ideals.
- **1946 (Coxeter, with Bruck).** Kirmse's J_1 is not closed (an explicit one-line witness); Bruck's remedy is a transposition of two imaginary basis units; the 21 such transpositions yield exactly the seven maximal orders; Coxeter's postscript supplies the missing connection to Dickson and Mahler.

The count converges over the four papers – $3 \to 8 \to 7$ – and the correction that turns Kirmse's J_1 into a genuine order is the *Coxeter–Bruck transposition* of two imaginary coordinate axes: a linear involution of $\mathbb{R}^8$.

This transposition is the historical precedent for the present paper's σ (Definition of the transposition-twisted product, Section 3). The two are the same kind of map – a coordinate transposition – differing only in what they are applied to. Coxeter–Bruck is applied to a maximal-order candidate that sits asymmetrically in $\mathbb{R}^8$, and so changes which E_8 placement is considered. The present σ is applied where the order is $L = D_8^+$, the coordinate-symmetric placement; hence $\sigma(L) = L$ and the E_8-level closure is preserved automatically ($L \cdot_\sigma L = \sigma(L \cdot L) \subseteq L$). The *Coxeter–Bruck transposition* thus appears below as the same operation at a different stratum: at the Leech-lattice level rather than at the maximal-order level, redirecting Wilson's sublattice conditions rather than repairing the choice of order.

References

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